

Exact Renewal and Clock Recovery for Odd-Affine Parity Maps on the 2-Adic Integers

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Abstract

For odd integers a, b , consider the parity map on the 2-adic integers that sends an even x to $x/2$ and an odd x to $(ax + b)/2$. Classical parity coding conjugates this family to the one-sided binary shift; we use that conjugacy as prior foundation, not as a novelty claim. We derive an exact first-return renewal description on the odd coset. The return time is $\tau(x) = v_2(ax + b)$ with conditional Haar law 2^{-k} , and the return map is a countable full shift with roof $r(k) = k$. Its ordinary Artin–Mazur series is unavailable because it has infinitely many fixed points, but the roof recovers the original clock: $(1 - \sum_{k \geq 1} z^k)^{-1} = (1 - z)/(1 - 2z)$, and restoring the zero orbit gives $(1 - 2z)^{-1}$. Moreover both the unweighted fixed counts and reciprocal 2-adic derivative stability sums are independent of every odd parameter pair. Exact solvability therefore supplies a parameter-blindness obstruction, not an arithmetic target model.

Keywords: 2-adic dynamics; parity coding; first return; renewal shift; dynamical zeta; Koopman isometry.

中文摘要

对任意奇整数 a, b ，本文研究二进整数环上的奇偶分支映射：偶点映为 $x/2$ ，奇点映为 $(ax + b)/2$ 。经典 parity 编码把该系统共轭到单边二元移位；本文把这一事实明确视为先验基础，而非新颖性主张。我们在奇数陪集上推导精确首返 renewal：返回时间为 $\tau(x) = v_2(ax + b)$ ，条件 Haar 分布为 2^{-k} ，首返映射是带屋顶 $r(k) = k$ 的可数字母满移位。加速映射因时间一已有无限多个不动点而不存在普通 Artin–Mazur 级数；但屋顶精确恢复原始时钟，再补回零轨道即得 $(1 - 2z)^{-1}$ 。同时，无权不动点计数与二进导数稳定性权重对所有奇参数都完全失明。因此，精确可解性给出的是参数盲目障碍，而非算术目标模型。

1 Frozen family and ownership boundary

Fix odd $a, b \in \mathbb{Z}$ and normalized Haar probability μ on $\mathbb{Z}_2$. The one-tick map is

$$T_{a,b}(x) = \begin{cases} x/2, & x \equiv 0 \pmod{2}, \\ (ax + b)/2, & x \equiv 1 \pmod{2}. \end{cases} \quad (1)$$

Oddness is structural: each numerator in the active branch is even. The parity vector $Q_{a,b}(x) = \sum_{j \geq 0} \epsilon_j(x) 2^j$, with $\epsilon_j(x) = T^j(x) \bmod 2$, conjugates $T_{a,b}$ to the dyadic shift. Bernstein and Lagarias treated this conjugacy and its odd $ax + b$ generality [1]. We claim no priority for it. The question here is what first-return acceleration does to clock and parameter information.

If $s_j = \sum_{i < j} \epsilon_i$, the inverse parity series is

$$Q_{a,b}^{-1} \left(\sum_{j \geq 0} \epsilon_j 2^j \right) = -b \sum_{j \geq 0} \epsilon_j 2^j a^{-s_{j+1}}. \quad (2)$$

It also shows that $Q_{a,b}$ preserves congruence cylinders and hence Haar measure. Multiplication by b conjugates $T_{a,1}$ to $T_{a,b}$.

2 Fixed words and stability blindness

For a word $\epsilon = (\epsilon_0, \dots, \epsilon_{n-1})$, put

$$A_\epsilon = \sum_{j=0}^{n-1} \epsilon_j 2^j a^{s_n - s_{j+1}}.$$

Branch induction gives

$$2^n T^n(x) = a^{s_n} x + b A_\epsilon. \quad (3)$$

Since $2^n - a^{s_n}$ is odd, every word has the unique fixed point

$$x_\epsilon = \frac{b A_\epsilon}{2^n - a^{s_n}} \in \mathbb{Z}_2. \quad (4)$$

The parity conjugacy makes all 2^n words admissible and distinct. Thus

$$\# \text{Fix}(T^n) = 2^n, \quad P(n) = \sum_{d|n} \mu(n/d) 2^d, \quad \zeta_{\text{AM}}(z) = \frac{1}{1-2z}. \quad (5)$$

Here $P(n)$ counts exact-period points, so $P(n)/n$ counts primitive cycles.

The branch derivatives are $1/2$ and $a/2$. On the point with word ϵ ,

$$(T^n)' = \frac{a^{s_n}}{2^n}, \quad \left| 1 - \frac{a^{s_n}}{2^n} \right|_2 = 2^n,$$

because $2^n - a^{s_n}$ is odd. Consequently

$$W_n := \sum_{x \in \text{Fix}(T^n)} |1 - (T^n)'(x)|_2^{-1} = 1, \quad \zeta_{\text{stab}}(z) = \frac{1}{1-z}. \quad (6)$$

Equations (5) and (6) are all-parameter theorems: both invariants erase a and b .

3 First-return renewal and clock recovery

Let $O = 1 + 2\mathbb{Z}_2$. For $x \in O$ with $ax + b \neq 0$, define

$$\tau(x) = v_2(ax + b), \quad R(x) = \frac{ax + b}{2^{\tau(x)}}. \quad (7)$$

The point $x_* = -b/a$ maps to zero and has infinite return time. More generally, let E be the set of odd points whose parity strings contain only finitely many ones. Under the parity homeomorphism, E is the union over all finite binary prefixes of one eventually-zero tail; it is therefore countable and Haar-null, and it contains x_* . Its complement $O_\infty = O \setminus E$ is invariant under successive returns and has full conditional Haar measure.

The affine map $x \mapsto ax + b$ sends O bijectively and measure-preservingly onto $2\mathbb{Z}_2$. Hence the normalized Haar layers give

$$\mu_O(\tau = k) = 2^{-k}, \quad k \geq 1. \quad (8)$$

The parity block associated with k is 10^{k-1} . Concatenation gives a bijection from O_∞ to $\{1, 2, \dots\}^{\mathbb{N}}$, conjugating R to the left shift. Bernoulli independence of disjoint parity blocks then makes successive returns iid with the geometric product measure. The constant block k has fixed point

$$x_k = \frac{b}{2^k - a}. \quad (9)$$

Thus R already has infinitely many time-one fixed points and its ordinary Artin–Mazur series is undefined.

Return symbol k represents k original ticks. With roof $r(k) = k$, the first-return series and renewal zeta are

$$F(z) = \sum_{k \geq 1} z^k = \frac{z}{1-z}, \quad \zeta_{\text{roof}}(z) = \frac{1}{1-F(z)} = \frac{1-z}{1-2z}. \quad (10)$$

Indeed $\log \zeta_{\text{roof}} = \sum_{n \geq 1} (2^n - 1)z^n/n$: these are exactly the original-clock periodic points whose binary words contain a one. The omitted all-zero fixed orbit contributes $(1-z)^{-1}$, and therefore

$$\zeta_{\text{roof}}(z)(1-z)^{-1} = \frac{1}{1-2z} = \zeta_{\text{AM}}(z). \quad (11)$$

4 Operator boundary and Route-A decision

On $H = L^2(\mathbb{Z}_2, \mu)$, $Uf = f \circ T$ is an isometry. It is not surjective: under parity coding, its range consists of functions independent of the first digit. The tail intersection $\bigcap_{n \geq 0} U^n H$ consists only of constants, while $H \ominus UH$ is an infinite-dimensional copy of the tail L^2 space. The Wold decomposition is therefore

$$U \simeq I_{\mathbb{C}} \oplus S^{(\aleph_0)}. \quad (12)$$

In particular $\sigma(U) = \overline{\mathbb{D}}$ and $\sigma_p(U) = \{1\}$. The isometry is noncompact and lies in no finite Schatten class, so no ordinary Fredholm determinant $\det(I - zU)$ is available. The two-sided inverse-limit shift is a same-clock unitary extension, but it changes the phase space and is not a repair of the frozen candidate. Koopman methods have also been studied for the discrete positive-integer $3x + 1$ system [2]; we make no general priority claim for a Collatz Koopman lift.

Gate	Exact verdict and blocking reason
A0_FAIL	Dyadic local arithmetic yields no rational-prime labels, prime-power repetition law, or von Mangoldt weight.
A1_WEAK	Primitive cycles are complete and reproducible, but have no intrinsic arithmetic labels or target amplitudes.
A2_FAIL	The three exact source zetas are elementary and parameter-blind; the accelerated map itself has infinitely many fixed points.
A3_FAIL	Rational continuation supplies no target functional equation, counting law, or natural Weil compression.
A4_FORMAL_HINT	The natural lift is a proper non-Schatten isometry; same-clock unitarization changes phase space.

Thus the tuple is (A0_FAIL, A1_WEAK, A2_FAIL, A3_FAIL, A4_FORMAL_HINT). A0 failure forces ROUTE_A_REJECTED; later-layer exactness cannot override the arithmetic entry gate.

5 Limitations and reproducibility

For $(a, b) = (3, 1)$ the 2-adic word 100 gives $1/5 \mapsto 4/5 \mapsto 2/5 \mapsto 1/5$. This legal $\mathbb{Z}_2$ cycle is not a positive-integer orbit, so the paper gives no progress on the positive-integer $3x + 1$ conjecture. If exactly one of a, b is even, the odd-branch numerator is odd and division by two leaves $\mathbb{Z}_2$; the theorem does not cross that boundary.

The release uses exact arithmetic only. A producer, independent checker, SymPy derivation, byte replay, semantic mutation suite, and content-addressed manifest are included. Finite ledgers are regression sentinels, not proofs. No prime table, target zero or divisor, arithmetic local factor, Euler factor, root number, automorphy, Hilbert–Pólya operator, Route-B authorization, external review, or acceptance rate is claimed. The scope literal is NO_BAD_EULER_OR_ROOT_NUMBER.

The final internal integrity pass separately cleared implementation bugs, citation identity and context, invented results, shortcut reliance, bug-as-insight reframing, methodology drift, and early-frame lock. This is an artifact-bounded audit, not a certificate of universal novelty or journal readiness. AI assistance was used for drafting and code generation; all claim-bearing formulas were reconstructed by independent exact code paths.

中文边界说明。 本文的精确结论限定于全体奇参数的 $\mathbb{Z}_2$ 动力学。经典 parity 共轭属于先验工作；有限回归表不替代理论证明； $(3, 1)$ 特例不推出正整数 Collatz 结论。由于 A0 缺少素数及素数幂来源，整体 Route A 必须拒绝。

References

- [1] D. J. Bernstein and J. C. Lagarias, “The $3x+1$ Conjugacy Map,” *Canadian Journal of Mathematics* 48 (1996), 1154–1169. doi:10.4153/CJM-1996-060-x.
- [2] J. Leventides and C. Poullos, “Koopman operators and the $3x + 1$ -dynamical system,” arXiv:2010.12987 (2020). arxiv.org/abs/2010.12987.